

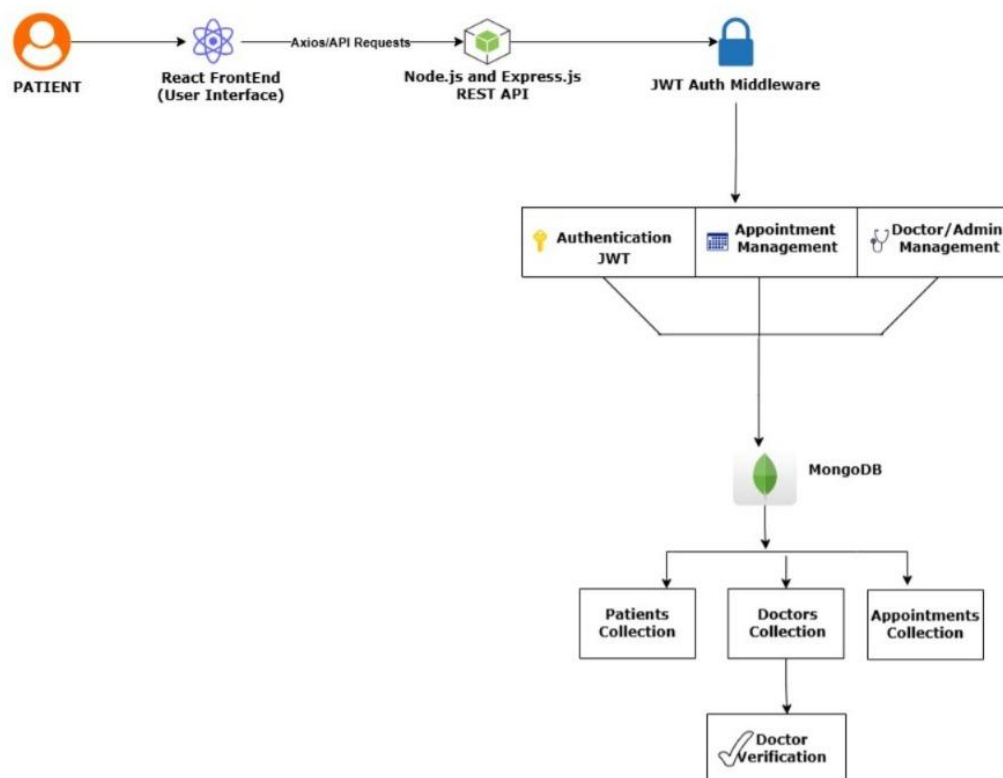
## Project Design Phase Solution Architecture

|               |               |
|---------------|---------------|
| Date          | 16 June 2026  |
| Team ID       | Srimai Gopala |
| Project Name  | Book A Doctor |
| Maximum Marks | 4 Marks       |

### Solution Architecture:

The Book-a-Doctor application utilizes a 3-layer MERN architecture which comprises of React, Express, and MongoDB. Three types of users can access the platform, namely Patients, Doctors, and Admins. Patients have access to searching for doctors and making appointments, Doctors are responsible for managing their appointments, and Admins oversee verification of doctors and all activities happening within the system. JWT is employed in the authentication process of accessing the application, with data stored in MongoDB.

### Solution Architecture Diagram:



The proposed Book-A-Doctor solution architecture uses a three-tier MERN stack along with MVC (Model View Controller) pattern for designing a clearly separated front-end, back-end, and database layers. Front-end made with React.js framework and powered by Vite is an implementation of View layer where users (Patients, Doctors, Administrators) can access and interact with the application using its interfaces.

Node.js & Express.js Rest API server serves as the controller and handles all the requests received from the client side using Axios HTTP requests. For security reasons, secured routes have been protected using JWT Authentication Middleware. Depending on the request, it routes the request further to the required controller module including AuthController, AppointmentController, DoctorController, AdminController, and UserController.

The controllers work with the database using Mongoose ODM to carry out CRUD operations within collections like User, Doctor, Appointment, and Audit Log. These collections constitute the model and determine the structure and relations within the application data. The database handles the requested data and sends back responses to the backend that are rendered by the frontend interface. Using the MVC-based MERN architecture, Book-A-Doctor application can be made highly secure and scalable while being able to manage healthcare appointments for patients, doctors, and admins effectively.